Exam Date & Time: 27-Nov-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (ELECTRONICS & COMMUNICATION ENGINEERING) EXAMINATIONS - NOVEMBER / DECEMBER 2024
SUBJECT: ECE 2125/ECE_2125 - ELECTROMAGNETIC WAVES

ELECTROMAGNETIC WAVES [ECE 2125]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

1A) Given points A (2, 5, -1), B (3, -2, 4) and C (-2, 3, 1). Find:
(i) $\bar{R}_{AB} \cdot \bar{R}_{AC}$
(ii) The angle between $\bar{R}_{AB}$ and $\bar{R}_{AC}$ (4)
(iii) The length of the projection of $\bar{R}_{AB}$ on $\bar{R}_{AC}$
(iv) The vector projection of $\bar{R}_{AB}$ on $\bar{R}_{AC}$

1B) Transform each of the following vectors to cylindrical coordinates at the specified point:
(i) 5 $\mathbf{a_x}$ at P ($\rho = 4$, $\phi = 120^0$, $z = -1$)
(ii) 5 $\mathbf{a_x}$ at Q ($x = 3$, $y = 4$, $z = -1$) (3)
(iii) 4 $\mathbf{a_x}$ - 2$\mathbf{a_y}$ - 4$\mathbf{a_z}$ at R ($x = 2$, $y = 3$, $z = 5$)

1C) An infinitely long uniform line charge is located at $y = 3$, $z = 5$. If $\rho_L = 30$ nC/m, find electric field intensity at
(i) The origin (3)
(ii) At P (0, 6, 1)

2A) Within the sphere r = 1 (m), the electric potential is given as $V = 10 + 10r^2 + \sin\theta + \cos\phi$ (V). Find the following at P ($r = 1$, $\theta = \pi/2$, $\phi = 0$):
(i) Electric potential
(ii) Electric field intensity
(iii) Electric flux density (4)
(iv) Volume charge density
(v) The charge that lies within the sphere r = 1 (m).

2B) Inside a cylindrical region, $\rho \leq 4$ (m), the electric flux density is given as $D = 5\rho^2\ \mathbf{a_\rho}$ (C/m^2). Find:
(i) The volume charge density at $\rho = 3$ (m)
(ii) The electric flux density $\rho = 3$ (m) (3)
The electric flux that leaves the cylinder $\rho = 3$, $|z| \leq 2.5$

2C) State and prove Gauss Divergence Theorem. (3)

3A) Let a filamentary current of 5 mA be directed from infinity to the origin on the positive z axis and then back out to infinity on the positive x axis. Find magnetic field intensity **H** at P (0, 1, 0). (4)

3B) If vector magnetic potential $\mathbf{A} = 10\rho^{1.5}\ \mathbf{a_z}$ Wb/m in free space, find:
(i) Magnetic field intensity **H**
(ii) Current density **J** (3)
(iii) Prove Ampere's Circuital Law for a circular path with $\rho = 1$.

3C) Derive the boundary conditions of static magnetic field at the interface of two different magnetic medium. (3)

4A) (4)

A lossy dielectric has an intrinsic impedance of $200\angle 30^0\,\Omega$ at a particular radian frequency ω. If, at that frequency, the plane wave propagating through the dielectric has the magnetic field component $H = 10e^{-ax}\cos\left(\omega t - \frac{1}{2}x\right)\hat{y}\ A/m$. Find E and α.

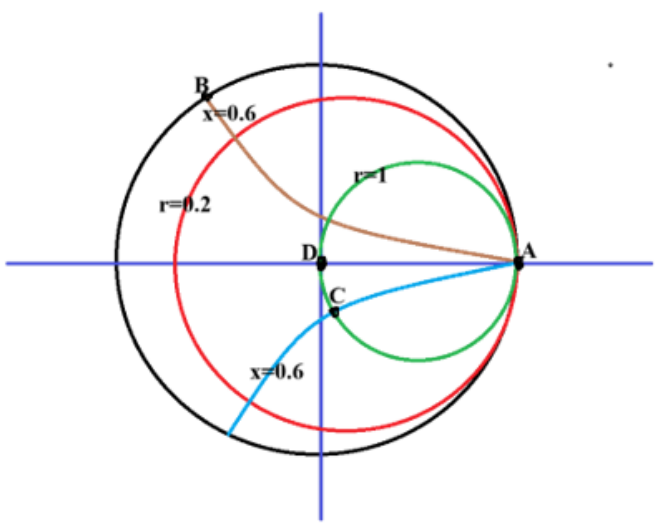
4B) In a non-magnetic medium $E = 4\sin\,(2\pi \times 10^7 t - 0.8x)\hat{z}$ V/m. Find (a) ε_r, η, (b) The time average power carried by the wave. (3)

4C) The electric field in free-space is given by $\vec{E}(z,t) = E_0^+\cos\,(2\pi * 10^6 t - \beta z)$. Using Maxwell's equations, find the magnetic field H. (3)

5A) A distortionless line has $Z_0 = 60\Omega$, $\alpha = 20\frac{mNp}{m}$, $u = 0.6c$, where c is the speed of light in a vacuum. Find R, L, G, C, and λ at 100 MHz. (4)

5B) A lossless transmission line with Z_0=50Ω is 30m long and operates at 2MHz. The line is terminated with a load Z_L=60+j40Ω. If velocity of wave = 0.6 C on the line, find, (i) the reflection coefficient, (ii) the standing wave ratio, (iii) the input impedance. Where C is velocity of light. (3)

5C)



(3)

Calculate the impedance at point A, B, C, and D.

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